

Challenge: Maze Solver

Points: 300 • Estimated Time: 90 Minutes

Challenge Scope

The robot must autonomously navigate from the entrance to the exit of a corrugated plastic classroom maze.

Instructions & Engineering Hints

- Welcome to the **Maze Solver Challenge**! Your mission is to program your robot to autonomously navigate from the designated entrance to the exit of the classroom maze without human intervention.
- **Strategic Approach:** Observe the geometry of the maze walls. Will your robot rely purely on reflex behaviors like wall-following, or will it make strategic decisions at intersections? Think about how state tracking or loop closures might help.
- **Sensor Placement Tips:** Consider where your sensors are mounted relative to the wheels and chassis. Placing distance or light sensors too far back can lead to clipping corners, while mounting them too far forward might make the robot overly sensitive to minor wall irregularities.
- **Structural Considerations:** Ensure your robot's chassis is narrow enough to comfortably fit inside the corridors with a safe margin for error. A stray wire or protruding sensor mount could easily get caught on the corrugated plastic walls.
- Test incrementally! Start by getting the robot to reliably drive down a single straight corridor before tackling 90-degree turns and complex dead ends.